

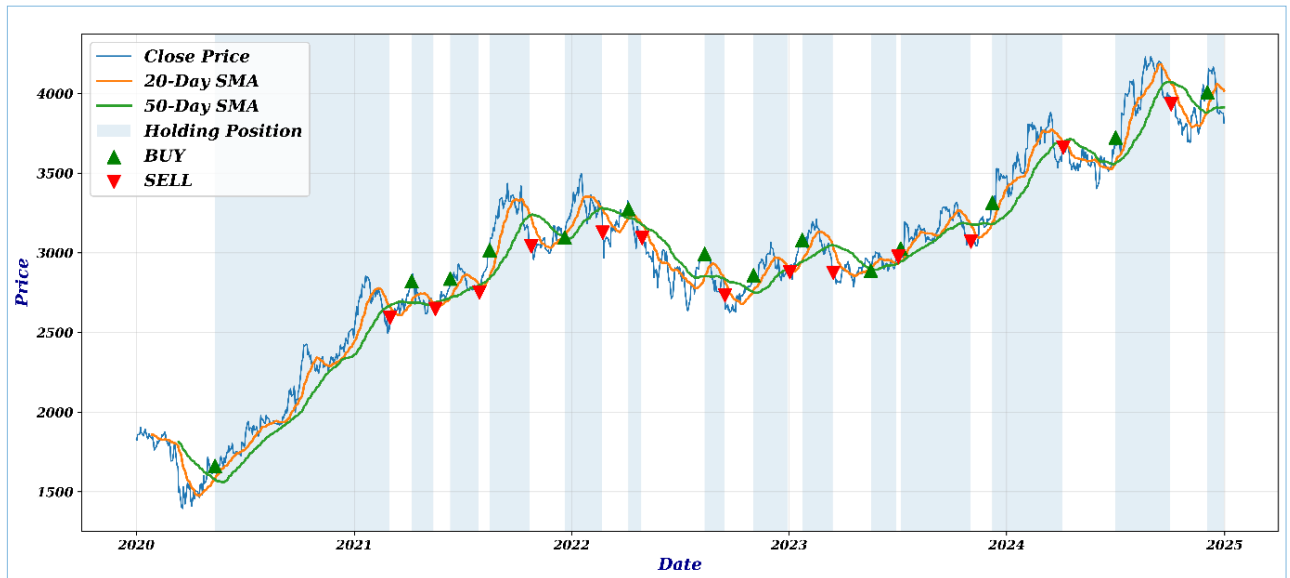


Figure 5: Buying, selling and holding stocks using the Moving Average Crossover Strategy

```
df["SMA50"],
label="50-Day SMA",
linewidth=2
)
# Holding periods
ax.fill_between(
    df.index,
    0,
    1,
    where=holding,
    transform=ax.get_xaxis_transform(),
    alpha=0.12,
    label="Holding Position"
)
# BUY
ax.scatter(
    df.index[buy],
    df.loc[buy, "Close"],
    marker="^",
    s=120,
    label="BUY",
    zorder=5
)
# SELL
ax.scatter(
    df.index[sell],
    df.loc[sell, "Close"],
    marker="v",
    s=120,
    label="SELL",
    zorder=5
)
ax.set_title(
```

```
f"{ticker} – Moving Average Crossover Strategy"
)
ax.set_xlabel("Date")
ax.set_ylabel("Price")
ax.legend()
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

Buy zone: In Figure 5, when the short moving average crosses over the long moving average, the stock should be bought.

Hold zone: When the short moving average stays over the long moving average, the stock should be held.

Sell zone: When the short moving average crosses under the long moving average, the stock should be sold.

This provides traders with a direct visual relationship between the crossover and the position, and between the position and the holding period.

Traders can optimise the entire process of SMA crossover through backtesting using different time periods of moving averages -- for example, 5/10, 10/30, and 50/200. However, algorithmic trading is not just about producing Buy and Sell signals. It needs much more than that to build a robust strategy. **END** 🐧

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The author is a technology enthusiast with interests in artificial intelligence, machine learning, natural language processing, and data-driven research. He has been involved in academic projects, technical writing, and training programmes that focus on the practical applications of AI.